have properties (which have no counterpart in everyday life) which make them appear sometimes as particles and sometimes as waves. This is closely related to the idea of quantum uncertainty.

the uncertainty principle

The importance of uncertainty in the quantum world was first spelled out by the German, Werner Heisenberg, in 1927. He realized that, in this context, uncertainty is a precise and definite thing, implicit in the equations that describe the behavior of things such as electrons. His analysis revealed that there are pairs of properties for

quantum entities, and it is impossible to specify precise values of both properties at the same time. The most important pair of these "conjugate variables" is position/momentum. His equations show that a particle ("quantum entity") cannot have a precisely determined position and a precisely determined momentum at the same time. And as momentum is proportional to velocity, this means that a particle does not have both position and velocity simultaneously.

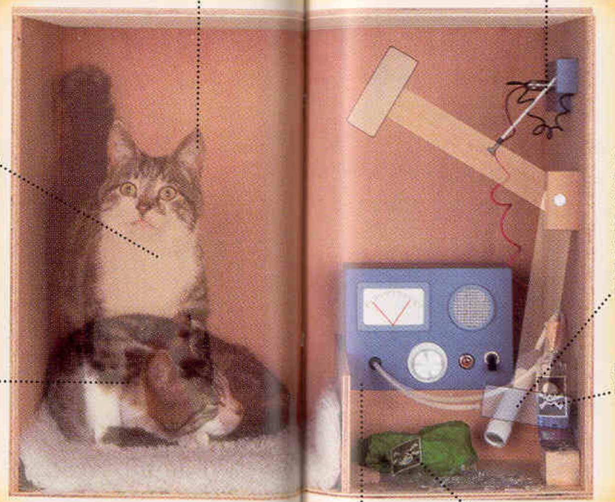
In fact, the uncertainty in position multiplied by the uncertainty in momentum is always greater than Planck's constant. We do not notice Planck's constant in everyday

Schrödinger's cat

Erwin Schrödinger dreamed up a thought experiment to demonstrate what he saw as the absurdity of quantum physics. He imagined a cat closed up in a windowless chamber with a plentiful supply of food and other creature comforts, but also with what he called a "diabolical device" in the form of a container of poison gas linked to a sample of radioactive material. If the radioactive material decays, the gas is released, and if the gas is released, the cat will die.

According to the quantum rules, it is possible to calculate a time when there is an exact 50/50 chance that the radioactive material has decayed and that the cat has survived or been killed. Quantum theory says that if we look into the box at that time, there will be a "collapse of the wave function" and we will see either a dead cat or a live cat. But it also says that if we don't look into the box, not just the radioactive sample, but the whole experiment, including the cat, is poised in a superposition of two possible states, each placed over the other, where the cat is both dead and alive at the same time, or half dead and half alive.

Schrödinger's thought experiment highlights the fact that there is some kind of boundary between the classical world of everyday objects and the quantum world of very small objects. Nobody quite knows where that boundary is, or how quantum effects disappear when that boundary is crossed.



is Schrödinger's cat dead or alive?
This mockup of how the "cat in a box" experiment might be done shows the cat and the diabolical device poised in a superposition of states that will only be resolved if the box is opened.